CellScope 5G v3.0

Comprehensive Technical Engineering Specification & System Architecture

Project Title: CellScope 5G — Cellular Network Intelligence, Passive Telemetry & Active QoE Analyzer
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Repository: https://github.com/TuhassinulArnob/cellCoverageApp.git (Branch: main)
Standard References: 3GPP TS 36.101 (E-UTRA RF), 3GPP TS 38.101 (5G NR User Equipment), 3GPP TS 45.005 (GSM/EDGE Radio), 3GPP TS 25.101 (UTRA FDD)

1. Executive Summary & Project Mission

CellScope 5G is a unified, multi-tier telecommunications diagnostic platform engineered for mobile network operators, RF field engineers, and academic researchers in information and communication technology (ICT). The system integrates an Android native field survey instrumentation agent, an asynchronous FastAPI ingestion and analytical microservice backend, and a high-performance React 18 / MapLibre geospatial survey intelligence dashboard.

The v3.0 milestone resolves critical operational requirements identified during extensive rural and urban drive-test passes along Bangladesh highway corridors (including the Kushtia–Bheramara and Islamic University survey tracks). Key engineering deliverables include: (1) automatic, background survey-triggered Active Quality of Experience (QoE) testing; (2) complete restoration of the visually stunning, bell-shaped multi-cell RF Frequency Channel Graph with multi-band auto-windowing and standard telecom dynamic limits; (3) multi-section, backward-compatible CSV audit engine; (4) precise 3GPP TS 45.005 GSM ARFCN band classification (P-GSM 900, E-GSM 900, GSM 850, DCS 1800); and (5) rigorous cellular traffic intelligence and frequency reuse modulo-3 diagnostics.

2. System Architecture & Tier Topology

The CellScope 5G ecosystem is decomposed into three highly cohesive, decoupled tiers operating across edge, cloud, and client layers:

Tier	Core Technologies	Key Responsibilities & Architectural Patterns
Mobile Edge (Android)	Kotlin 2.0, Android 14/15 APIs, Jetpack Room SQLite, Coroutines / Flow, OkHttp 4, Material Design 3	Low-latency telephony signal extraction (TelephonyManager), continuous GPS trajectory recording, background survey foreground services, automated active HTTP/TCP QoE testing loop, on-canvas custom RF spectrum rendering, multi-section CSV/PDF export.
Backend Analytics	Python 3.14, FastAPI 0.136, Pydantic v2, SQLAlchemy 2.0, Uvicorn, Pytest	High-throughput RESTful ingestion endpoints, multi-section survey CSV parsing and schema validation, geospatial spatial clustering, hexagonal coverage estimation, and AI radio quality anomaly diagnostics.
Web GIS Dashboard	React 18, Vite 8, TypeScript, MapLibre GL, Tailwind CSS, Lucide Icons, Chart.js	Interactive geospatial heatmap rendering, multi-layer layer visibility toggling, serving vs neighbor spatial inspection, cell tower triangulation rendering, and temporal drive-test playback.

3. RF Channel Frequency Graph Restoration & Visualization Engine

In previous iterations, the RF Frequency Graph suffered from a blank rendering defect when observed cells spanned multiple distant frequency bands (e.g. Band 8 at 940 MHz and Band 3 at 1850 MHz). In addition, an arithmetic operator precedence bug in carrier channel boundary calculation ((fCenter - (bwMhz / 2.0) * 10.0).toInt() / 10.0) erroneously computed channel lower frequencies as 180.0 MHz instead of 1845.0 MHz. This caused the plot viewport to span over 1,600 MHz, compressing legitimate 5–20 MHz channels into imperceptible sub-pixel slivers.

Architectural Corrections & Enhancements:
• **Mathematical Channel Envelope Model:** Each cellular channel is visualized as a continuous, smooth bell-shaped curve derived from its 3GPP nominal channel bandwidth and center frequency using two chained cubic Bézier curves:

$$P_0 = (x_{start}, y_{base}), \quad P_1 = (x_{start} + 0.35\cdot \Delta x, y_{base}), \quad P_2 = (x_{center} - 0.35\cdot \Delta x, y_{peak}), \quad P_3 = (x_{center}, y_{peak})$$

$$P_4 = (x_{center} + 0.35\cdot \Delta x, y_{peak}), \quad P_5 = (x_{end} - 0.35\cdot \Delta x, y_{base}), \quad P_6 = (x_{end}, y_{base})$$

• **Multi-Band Smart Auto-Focus Windowing:** When "All Bands" is active and the aggregate carrier span across detected cells exceeds 60.0 MHz, the graph automatically windows onto the authoritative serving cell's operational band (or the modal band with greatest observation density), with a non-intrusive notification indicator ("Auto-focused: Band X • Filter bands above"). Users can instantly switch to any specific band (B1, B3, B8, B28, B40) using dedicated horizontal filter chips.
• **Minimum Visual Half-Width Safeguard:** For narrow channels such as GSM 200 kHz or LTE 1.4 MHz, a display constraint enforces halfWidthPx = max(nominalHalfWidthPx, 16f * density), guaranteeing that every active channel renders a clear, legible envelope regardless of screen pixel density or zoom factor.
• **Standard Telecom Dynamic Range:** The Y-axis adopts fixed, telecom-standard reference levels from **-140 dBm to -40 dBm** with equidistant 20 dB gridlines (-140, -120, -100, -80, -60, -40 dBm), eliminating vertical viewport jitter.
• **Visual Role Differentiation:** Serving cells are stroked with primary blue (#0284C7, stroke width 2.8dp) and filled with 21.5% alpha, featuring a 4.5dp center peak marker. Unconfirmed neighbor cells are rendered in muted slate (#64748B, stroke width 1.5dp, 7.8% alpha).
• **Touch Hit-Testing & Inspection Panel:** Touching any envelope or peak opens an amber glowing ring (#F59E0B) and callout pill, simultaneously updating the *Cell Carrier Inspection* card below with duplex separation, bandwidth, RSRP, RSRQ, and SINR.
• **Explicit 3GPP Disclaimer:** Graph headers clearly state: *"Visualized Channel Signal Distribution (Derived from 3GPP channel allocation; not calibrated spectral power)"*.

4. 3GPP Standardization & Carrier Resolution Formulations

CellScope 5G adheres strictly to 3GPP specifications for all carrier center frequency and channel edge calculations. Raw channel numbers extracted from Android telephony APIs are transformed through the following standardized mathematical formulas:

Radio Access Tech (RAT)	3GPP Standard	Center Frequency Formulation (Downlink)	Duplex / Band Range
LTE (4G)	3GPP TS 36.101	$F_{DL} = F_{DL_low} + 0.1 \cdot (N_{DL} - N_{Offs_DL})$	B1, B3, B8, B28 (FDD); B38, B40, B41 (TDD). Duplex sep: $\|F_{DL} - F_{UL}\|$
5G NR (Sub-6 GHz)	3GPP TS 38.101-1	0-3 GHz: $F_{REF} = \Delta F_{Global} \cdot N_{REF}$ (5 kHz) 3-24 GHz: $3000 + 0.015 \cdot (N_{REF} - 600000)$	n1, n3, n8, n28 (FDD); n40, n41, n77, n78 (TDD). 15/30 kHz SCS.
GSM / 2G	3GPP TS 45.005	P-GSM (1..124): $935.0 + 0.2 \cdot n$ E-GSM (975..1023): $925.0 + 0.2 \cdot (n - 1024)$ DCS 1800 (512..885): $1805.2 + 0.2 \cdot (n - 512)$	FDD: 45 MHz duplex sep (900 MHz), 95 MHz sep (1800 MHz), 45 MHz sep (850 MHz).
WCDMA (3G)	3GPP TS 25.101	$F_{DL} = 0.2 \cdot UARFCN$	B1 (2100 MHz, 190 MHz sep), B8 (900 MHz, 45 MHz sep). 5.0 MHz BW.

5. Mathematical Models & Radio Analytics

5.1 LTE Radio Signal Metric Interrelationships

In LTE networks, the relationship between Reference Signal Received Power (RSRP), Reference Signal Received Quality (RSRQ), and Carrier Received Signal Strength Indicator (RSSI) across N Resource Blocks (RBs) is mathematically defined as:

$$\text{RSRQ} = \frac{N \cdot \text{RSRP}}{\text{RSSI}} \implies \text{RSRQ}_{\text{dB}} = 10\log_{10}(N) + \text{RSRP}_{\text{dBm}} - \text{RSSI}_{\text{dBm}}$$

5.2 LTE Cell-Specific Reference Signal (CRS) Modulo-3 Frequency Shift Grouping

To mitigate downlink intra-frequency pilot subcarrier collision between adjacent sectors, LTE applies a subcarrier shift to the reference signals based on the Physical Cell Identity (PCI, $N_{\text{ID}}^{\text{cell}} \in [0, 503]$):

$$v_{\text{shift}} = N_{\text{ID}}^{\text{cell}} \bmod 3 = (3 \cdot N_{\text{ID}}^{(1)} + N_{\text{ID}}^{(2)}) \bmod 3 = N_{\text{ID}}^{(2)} \in \{0, 1, 2\}$$

Engineering rule: Adjacent cells sharing identical carrier frequency and identical v_shift suffer severe CRS puncture interference, degrading PDSCH decoding even under strong RSRP. CellScope 5G isolates and flags CRS collisions exclusively for LTE networks (excluding 5G NR and 2G GSM).

5.3 Cellular Frequency Reuse Distance & Empirical Radius

For a cellular cluster with reuse factor N, the theoretical co-channel reuse distance D as a function of cell radius R is:

$$D = R \cdot \sqrt{3N}, \quad Q = \frac{D}{R} = \sqrt{3N} \quad (\text{Co-channel reuse ratio})$$

Where cell radius R is empirically calculated from GPS drive trajectory coordinates:

$$R_{\text{emp}} = \max_{i} \left\{ \mathcal{H}(\phi_{\text{est}}, \lambda_{\text{est}}, \phi_i, \lambda_i) \right\}, \quad \mathcal{H}(\phi_1, \lambda_1, \phi_2, \lambda_2) = 2R_{\text{Earth}} \arcsin \sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1\cos\phi_2\sin^2\left(\frac{\Delta\lambda}{2}\right)}$$

6. Passive Radio Telemetry & Cellular State Machine

The edge instrumentation engine captures raw physical-layer telephony data without active transmission overhead. Key architectural components include:
• **Authoritative Serving Cell Classification:** Rather than assuming the first listed cell or lowest PCI is the serving sector, ServingCellClassifier.kt evaluates multiple independent evidence fields: registered state (isRegistered = true), connected radio status (CELL_CONNECTION_PRIMARY_SERVING), Cell Identity (ECI/CI/NCI presence), and positive timing advance. Unconfirmed neighbors are strictly separated from primary serving carriers.
• **5G NR NSA vs. SA Telemetry:** Operates with Android 11–15 TelephonyDisplayInfo overrides, distinguishing OVERRIDE_NETWORK_TYPE_NR_NSA (Non-Standalone with LTE anchor) from OVERRIDE_NETWORK_TYPE_NR_ADVANCED (true NR carrier aggregation / mmWave) and true 5G SA standalone registration.
• **Multi-SIM DSDS Support:** Actively monitors subscription IDs, binding telemetry samples to specific operator slots (e.g. Grameenphone vs. Banglalink).

7. Automated Quality of Experience (QoE) Engine & Survey Lifecycle

A signature upgrade of CellScope 5G v3.0 is the complete automation of active network QoE testing tied directly to survey recording:
• **Zero-Touch Activation:** When the user presses *Start Survey*, the active QoE scheduler automatically triggers an initial bounded measurement (download throughput, upload throughput, RTT, jitter, and packet loss) as soon as endpoints are resolved.
• **Configurable Periodic Intervals:** Supports 5-minute, 10-minute (default), and custom scheduling intervals. Periodic tests execute within an Android Foreground Service (QoeForegroundService) with required foreground notification channels, preventing Doze mode termination.
• **Elimination of Arbitrary Caps:** Removed the previous hardcoded 50 MB daily cap, replacing it with user-configurable per-test payload size limits, cumulative survey data caps, and safe cellular roaming safeguards.
• **Lifecycle Cleanliness:** Pausing a survey gracefully halts the timer; resuming schedules the next bounded cycle without overlapping workers; stopping the survey cleanly terminates workers while guaranteeing that all completed tests remain persisted in Room SQLite.

8. Multi-Scale Cellular Traffic Intelligence & Congestion Diagnostics

CellScope 5G provides real-time, evidence-based congestion detection through TrafficIntelligenceEngine.kt and CongestionEvidenceEngine.kt:
• **Evidence-Based Congestion Verdicts:** Identifies sector saturation when measured RSRP is strong (> -90 dBm) but RSRQ is poor (< -14 dB) and active download throughput drops below 2.0 Mbps.
• **Stronger Neighbor Intelligence ("Why Not Connected?"):** Compares serving cell RSRP against neighbor cells on identical or inter-frequency carriers. If a neighbor is observed with $\Delta \text{RSRP} \ge +6.0\text{ dB}$ for more than 5 consecutive observation windows, the engine classifies the anomaly (e.g. Handover Ping-Pong, Cell Individual Offset Bias, or Missing Neighbor Relation).

9. Multi-Section CSV Engine & Schema Integrity

Field survey exports frequently contain heterogeneous data streams. Rather than relying on fragile single-table CSVs, CellScope 5G v3.0 establishes an audited 6-section composite CSV format:
1. [SURVEY_METADATA] — Session ID, operator, device model, Android SDK, survey start/end timestamps.
2. [TELEMETRY_SAMPLES] — Full continuous passive radio measurements (Timestamp, Lat, Lon, Speed, RAT, Band, ARFCN, PCI, RSRP, RSRQ, SINR).
3. [ACTIVE_QOE_TESTS] — Bound HTTP/TCP test results (Test ID, Direction, Throughput Mbps, RTT Mean/P95, Jitter, Probe Loss).
4. [MOBILITY_EVENTS] — Handover transitions, cell reselection events, RAT inter-system handovers, and drop anomalies.
5. [SERVICE_HOLES] — Continuous coverage hole stretches where signal fell below -115 dBm or Out-of-Service occurred.
6. [ESTIMATED_TOWERS] — Triangulated eNodeB/gNodeB coordinates with confidence radii.

CsvImporter.kt parses every section independently with schema validation, reporting malformed row numbers without discarding legitimate telemetry samples.

10. Geospatial Mapping & GIS Visualization Engine

The frontend GIS console integrates MapLibre GL with custom vector tiles, providing real-time trajectory visualization:
• **Dynamic Layer Toggle:** Independent layer filtering for LTE, 5G NR, 2G GSM, Coverage Holes, and Estimated Towers.
• **Viewport State Preservation:** Zoom level, center coordinates, and active layer selections are persisted across theme toggles (Light/Dark mode) and screen orientation changes.
• **Coverage Hole Buffer Polygons:** Automatically outlines service hole stretches with GPS jitter filtering and speed interpolation.

11. Applied ICT & Telecom Engineering Academic Foundations

This project directly translates theoretical ICT and telecommunications engineering coursework principles into industrial-grade software:
• **Cellular Mobile Communications:** Cellular concept, frequency reuse clusters ($N=3, 7, 12$), co-channel interference reduction factor (Q), and adjacent-channel interference rejection.
• **Radio Frequency Propagation Models:** Log-distance path loss, Okumura-Hata empirical models, and shadow fading log-normal distributions.
• **Digital Signal Processing & Modulation:** OFDM subcarrier allocation, Cyclic Prefix duration, QPSK/16QAM/64QAM/256QAM constellation mapping, and reference signal pilot symbol placement.
• **Computer Networks & Queuing Theory:** TCP congestion control dynamics (BBR/CUBIC), bufferbloat detection, RTT variance (Jacobson's algorithm), and jitter formulation (RFC 3550).

12. Backend Architecture & Web GIS Frontend Stack

• **FastAPI Microservice (Backend):** Fully asynchronous REST API (Uvicorn / AnyIO) with Pydantic v2 data models for rigorous schema enforcement. Endpoints support survey session ingestion (POST /api/v1/surveys/upload), analytical report generation, and geojson export.
• **React 18 / Vite 8 Dashboard (Frontend):** Modern TypeScript client utilizing Tailwind CSS and Lucide icons. High-performance canvas layers render up to 50,000 drive-test points without DOM throttling.

13. Comprehensive Verification & Test Matrix

Every module within the CellScope 5G platform is subjected to automated regression and unit verification:

Test Suite / Component	Scope & Validated Assertions	Test Count	Result
Android Unit Tests (testDebugUnitTest)	52 test suites: CellularBandResolver, ServingCellClassification, RfCarrierVisualization, GsmArfcnClassification, PciReuseAnalysis, MultiSectionCsvRegression, QoeAutomation, DynamicMapAndCoverage.	272 tests	**100% PASSED (0 failures)**
Android APK Build (assembleDebug)	Full compilation, KSP symbol processing, data-binding code-generation, resource packaging, and DEX assembly.	app-debug.apk (~49.8 MB)	**BUILD SUCCESSFUL**
FastAPI Pytest Suite (backend/tests)	RESTful endpoints, RF mathematical formulas, Pydantic schema validation, and CSV export serialization.	19 tests	**100% PASSED (0 failures)**
Frontend Production Build (npm run build)	TypeScript type-checking (tsc -b) and Vite 8 bundle production optimization (1,932 modules).	dist/ bundle	**BUILT (0 errors)**
Multi-Section CSV Regression (MultiSectionCsvRegressionTest)	Validates ingestion of genuine 6-section highway survey dataset (cellscope_extended_Survey_...csv).	End-to-End fixture	**100% VERIFIED**

14. Build Instructions & Environment Setup

Android Application:
Prerequisites: JDK 21+ (e.g. Android Studio JBR at C:\Program Files\Android\Android Studio\jbr), Android SDK 35.
Command to test: $env:JAVA_HOME = "C:\Program Files\Android\Android Studio\jbr"; .\android\gradlew.bat -p android testDebugUnitTest
Command to build APK: $env:JAVA_HOME = "C:\Program Files\Android\Android Studio\jbr"; .\android\gradlew.bat -p android assembleDebug

FastAPI Backend:
Prerequisites: Python 3.11+, virtualenv.
Run tests: python -m pytest backend/tests
Run dev server: uvicorn backend.app.main:app --host 0.0.0.0 --port 8000 --reload

React GIS Frontend:
Prerequisites: Node.js 18+, npm.
Run build: npm --prefix frontend run build
Run dev server: npm --prefix frontend run dev

15. References & Regulatory Standards

1. **3GPP TS 36.101:** Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception.
2. **3GPP TS 38.101-1 / TS 38.101-2:** NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone, Part 2: Range 2.
3. **3GPP TS 45.005:** Radio Access Network; GSM/EDGE Radio transmission and reception.
4. **3GPP TS 25.101:** Universal Mobile Telecommunications System (UMTS); User Equipment (UE) radio transmission and reception (FDD).
5. **IETF RFC 3550:** RTP: A Transport Protocol for Real-Time Applications (Interarrival Jitter Calculation).
6. **IETF RFC 6349:** Framework for TCP Throughput Testing.

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